INFRASTRUCTURE

Active Directory Documentation

- Active Directory Components
- Active Directory Replication Technologies
- Organizational Unit (OU) Structure
- Active Directory Trust
- Group Policy
- LennarCorp DNS Infrastructure

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VERSION HISTORY

Version	Date	Name	Comments/Changes
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DOCUMENT APPROVER

Document Approver	Designation	Signature	Approval Date
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ABBREVIATIONS AND ACRONYMS

Term	Description
AD DS	Active Directory Domain Services
DC	Domain Controller
DFS	Distributed File System
DFSR	Distributed File System Replication
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
FRS	File Replication Service
FSMO	Flexible Single Master Operation
GPMC	Group Policy Management Console
GPO	Group Policy Objects
NIC	Network Interface Card
OU	Organizational Units
PDC	Primary Domain Controller
RID Master	Relative ID Master
RSoP	Resultant Set of Policy
SOM	Scope of Management
WMI	Windows Management Instrumentation

OVERVIEW

Lennar has an AD forest that consists of one child domain. Lennarcorp.com is the parent domain and lennar.lennarcorp.com is the child domain where all the users reside for directory services.

The Lennarcorp domain consists of 6 domain controllers with MDC-LC-DC01 as the FSMO role holder. The Lennar.Lennarcorp.com domain consists of 10 domain controllers with MDC-DC01 as the FSMO role holder. The Forest Function level is Windows Server 2008 R2 and the domain level for each domain is Windows Server 2008 R2.

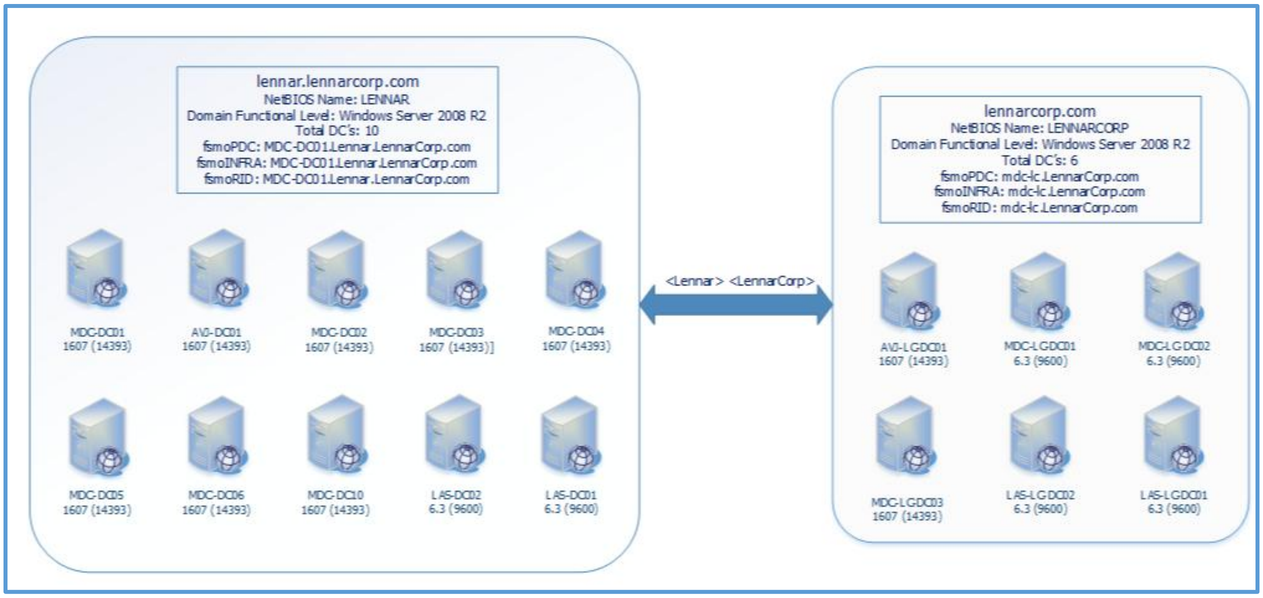

Figure 1: Lennar's Active Directory Environment

AUDIENCE

The intended audience of this document are the Lennar Technology Solutions (LTS) personnel.

PREREQUISITE

Before you read this document, you should have a good understanding of how AD DS works on a functional level. You should understand the organizational requirements that will be reflected in your AD DS deployment strategy.

ACTIVE DIRECTORY COMPONENTS

DOMAIN CONTROLLER

A domain controller is a server that runs a version of the Windows Server operating system and has Active Directory Domain Services installed. By default, a domain controller stores one domain directory partition consisting of information about the domain in which it is located, plus the schema and configuration directory partitions for the entire forest.

FSMO roles in Active Directory

Where do these servers reside?

The domain controllers that hold the FSMO roles reside in the Miami Datacenter.

How are they structured?

There are 5 FSMO roles. 2 - Forest Wide and 3 - Domain Wide. The FSMO roles reside in one server per domain.

Lennarcorp.com

Term	Description
Forest: Schema master	MDC-LC-DC01.LennarCorp.com
Forest: Domain naming master	MDC-LC-DC01.LennarCorp.com
Domain: PDC	MDC-LC-DC01.LennarCorp.com
Domain: RID pool manager	MDC-LC-DC01.LennarCorp.com
Domain: Infrastructure master	MDC-LC-DC01.LennarCorp.com

Lennar.Lennarcorp.com

Term	Description
Domain: PDC	MDC-DC01.Lennar.LennarCorp.com
Domain: RID pool manager	MDC-DC01.Lennar.LennarCorp.com
Domain: Infrastructure master	MDC-DC01.Lennar.LennarCorp.com

Microsoft Recommendation:

- *The following roles to be located on the server holding the Forest PDC Role: Schema Master Role and Domain Naming Master Role.*
- *The following role to be hosted on the domain PDC role holder: RID Master Role. The Infrastructure Master Role is not used when all domain controllers in the domain are global catalog servers. This role can be hosted on the same server holding the PDC role in the domain.*

Microsoft Article: https://support.microsoft.com/en-us/help/223346/fsmo-placement-and-optimization-on-active-directory-domain-controllers

REPLICATION ENGINE

Distributed File System Replication (DFSR)

DFS Replication is a role service in Windows Server that enables you to efficiently replicate folders (including those referred to by a DFS namespace path) across multiple servers and sites. DFS Replication is an efficient, multiple-master replication engine that you can use to keep folders synchronized between servers across limited bandwidth network connections. It replaces the File Replication Service (FRS) as the replication engine for DFS Namespaces, as well as for replicating the Active Directory Domain Services (AD DS) SYSVOL folder in domains that use the Windows Server 2008 or later domain functional level.

What is the replication interval?

Replication interval is every 15 minutes.

How are we replicating?

Active Directory uses Knowledge Consistency Checker (KCC) to replicate between domain controllers. It creates the topology and calculates where to create replication links between domain controllers and determines the most efficient way for replication to work.

LENNARCORP DNS INFRASTRUCTURE

OVERVIEW

The *LennarCorp.com* namespace is the AD Forest and Secure Root Domain. There is one child domain, *Lennar.lennarcorp.com* which fall under this forest. In addition, there are 9 other forests with connectivity via trusts and various DNS mechanisms.

All DNS servers are a role on a Domain Controller. There are 21 in the forest which include 6 in the forest root lennarcorp.com and the remaining 15 in the *Lennar.lennarcorp.com* domain.

SERVERS DNS CONFIGURATION

The following standardized configuration will be applied to the 7 division DCs.

NIC DNS Order

1. MDC-DC03
2. MDC-DC02
3. The servers own private IP (Not the 127.0.0.1 loopback) – If server is a domain controller.

Private DNS Forwarders

1. MDC-LC-DC02
2. MDC-LC-DC01
3. AVJ-LC-DC01

DNS Zones: Only the *Lennar.lennarcorp.com* AD-Integrated zone will be hosted.

Dynamic Updates: Secure Only

Zone Transfers: Disabled/None

AD CONDITIONAL FORWARDERS

There are currently 22 conditional forwarders configured to replicate through the DCs. The use of conditional forwarders provides several advantages including ease of management and updates and increased security.
Here is the current list of conditional forwarders:

- calatl.com
- calatlcorp.com
- CORP.CALATL.COM
- corp.flrealtycorp.com
- corp.wci.local
- dai.local
- eaglehm.com
- eaglehmcorp.com
- ehm.com
- empowerplan.com
- kymc.com
- lenlab.test
- mdcdmz.com
- nat.com
- pinnacle-mortgage.com
- RYLAND.COM
- STANDPAC.com
- stanpac.com
- uamc.com
- uamcglobal.com
- wcicommunities.com

DNS CONFIGURATION – DATACENTER DOMAIN CONTROLLERS

The following configuration will be applied to the five domain controllers located in the Miami Datacenter location.

DNS Order

1. MDC-DC02 (except on MDC-DC02 which will use MDC-DC01)
2. MDC-DC03 (except on MDC-DC03 which will use MDC-DC01)
3. The servers own private IP (Not the 127.0.0.1 loopback)

Private DNS Forwarders

1. MDC-LC-DC02
2. MDC-LC-DC01
3. AVJ-LC-DC01

DNS Zones: Only the lennar.lennarcorp.com AD-Integrated zone will be hosted

Dynamic Updates: Secure Only

Zone Transfers: MDC-DC01/02/03 will allow zone transfers to specified DNS servers

LENNARCORP.COM MDC DOMAIN CONTROLLERS

The following configuration will be applied to the three domain controllers located in the Miami Datacenter location. Public DNS forwards are only applied on the following servers: MDC-LC-DC01, MDC-LC-DC02 and AVJ-LC-DC01. The rest will point to the domain controllers that are configured with public DNS forwarders.

DNS Order

1. MDC-LC-DC02 (except on MDC-LC-DC02 which will use MDC-LC-DC03)
2. MDC-LC-DC01 (except on MDC-LC-DC01 which will use MDC-LC-DC03)
3. The servers own private IP (Not the 127.0.0.1 loopback)

Public DNS Forwarders

1. 12.127.16.67 - ATT.net Public DNS
2. 64.6.64.6 - Verisign/UltraDNS Public DNS
3. 64.6.65.6 - Secondary Verisign/UltraDNS Public DNS

Private DNS Forwarders

1. MDC-LC-DC02
2. MDC-LC-DC01
3. AVJ-LC-DC01

The following figure shows how clients will send Forward and Reverse DNS queries to the DNS servers hosted in the Miami Datacenter. Internal DNS queries for Lennar.Lennarcorp.com will be resolved by the internal DNS servers along with any queries for other domains.

Queries for public DNS will be forwarded over to the Lennarcorp.com DNS servers that have Public DNS forwarders configured to forward the requests to public hosted DNS servers and forward the response to the clients.

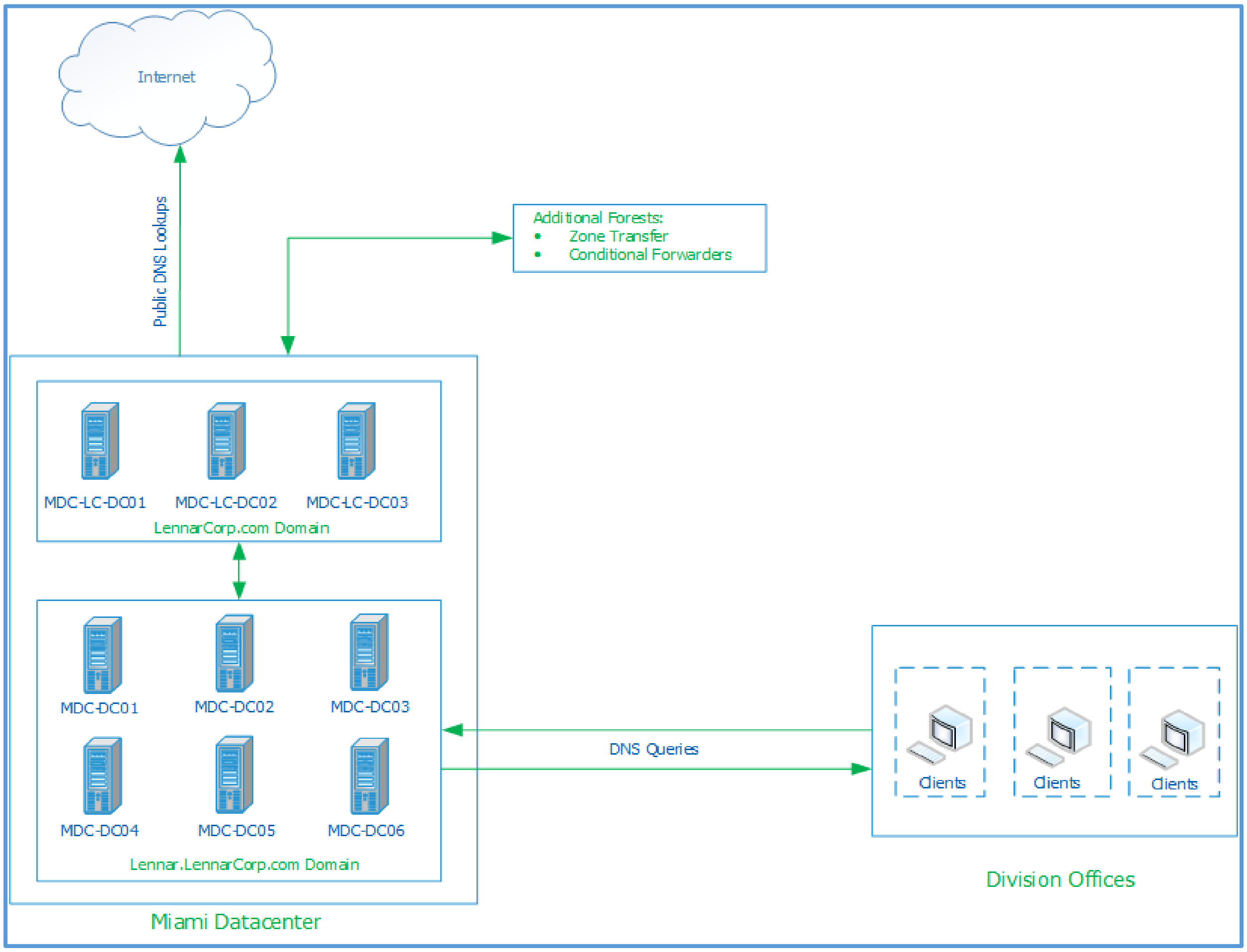

Figure 2: Lennar DNS Infrastructure Overview

ACTIVE DIRECTORY TRUST

An Active Directory trust is a logical link that allows one domain to access another domain, or a forest to access another forest. Active Directory Trusts makes it possible to grant access to resources in a trusted domain or forest like users, groups and computers.
Lennarcorp.com and *Lennar.Lennarcorp.com* utilizes different types of Active Directory trusts to different forests. These trusts allow different forests and domains use resources hosted in the Lennar environment and vice versa.

Forest Trust: A trust that is explicitly transitive between two forest root domains. The Forest Trust can be one-way or two-way. An example of a transitive trust is *calatlcorp.com*.

List of established Trusts

- Domain: Lennarcorp.com
- Forest Trust:

  calatlcorp.com

  ehm.com

  mdcdmz.com

  nat.com

  uamc.com

- Domain: Lennar.Lennarcorp.com
- Two-Way Trust:

  calatlcorp.com

  corp.calatl.com

  corp.flrealtycorp.com

  eaglehmcorp.com

  ehm.com

  mdcdmz.com

  nat.com

  ryland.com

  standpac.com

  uamc.com

- One-Way Trust:

  corp.wci.local

The following figure shows the connected trusts that Lennarcorp.com forest has with other forests to access resources hosted in the other domains and in the Lennarcorp.com domains.

Lennarcorp.com has a two way trust with the child domain lennar.lennarcorp.com and some essential core domains such as eaglehmcorp.com and calatlcorp.com for forest level resources such as Active Directory Certificate Services.

Lennar.Lennarcorp.com has two-way and one-way trusts between other Active Directory domains in order to share domain resources such as File Sharing Services, Legacy applications and current Active Directory Integrated applications.

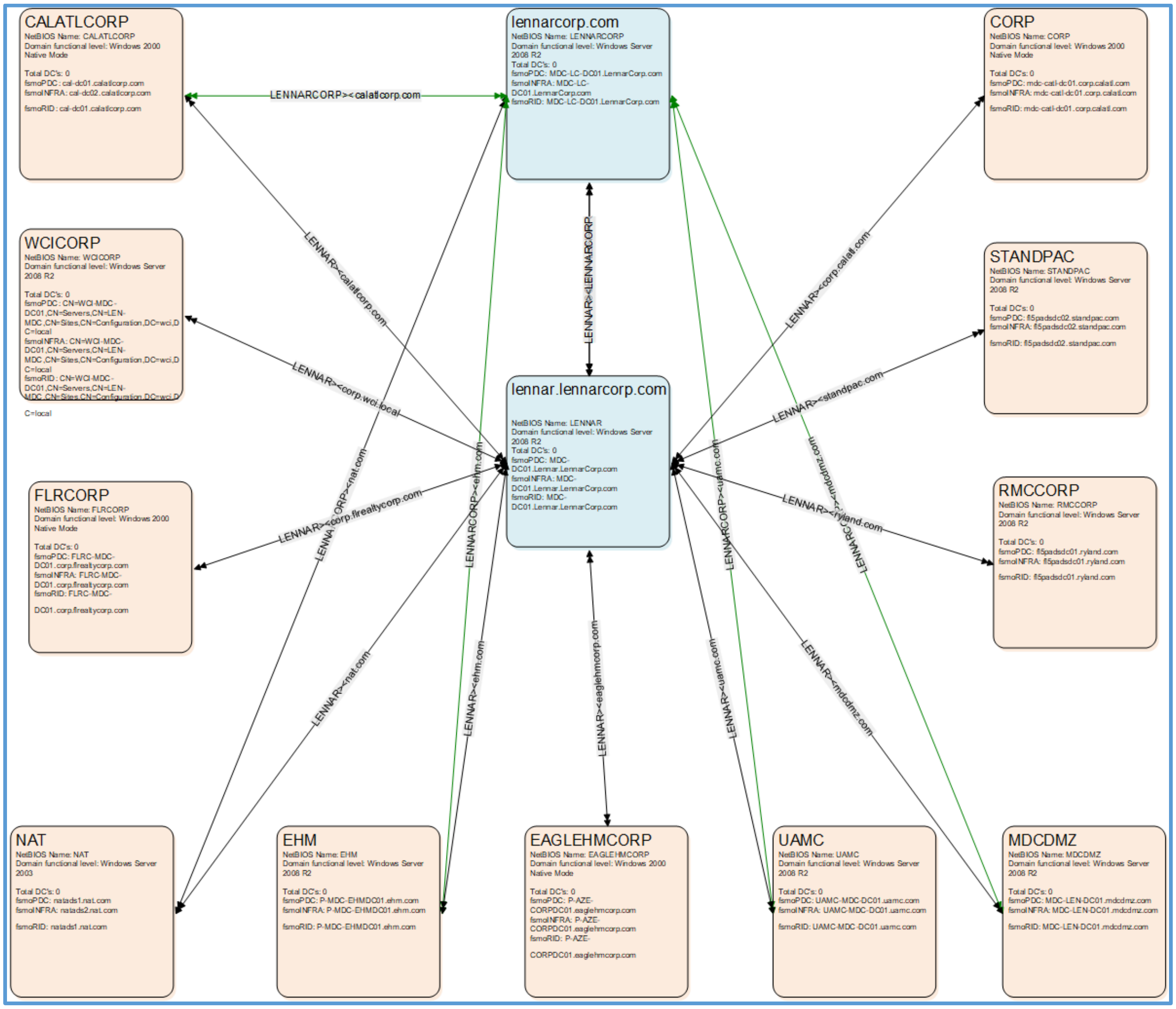

Figure 3: Active Directory Trusts

ACTIVE DIRECTORY REPLICATION TECHNOLOGIES

A directory service site topology is a logical representation of the physical network. Designing a site topology for Active Directory Domain Services (AD DS) involves planning for domain controller placement and designing sites, subnets, site links, and site link bridges to ensure efficient routing of query and replication traffic.

Intrasite Replication Topology

Intra-site replication takes place between domain controllers within the same site, making it a fairly uncomplicated process. When changes are made to the replica of Active Directory on one particular domain controller, the domain controller contacts the other domain controllers within the same site and it then checks the information it contains against information hosted by the other domain controllers.

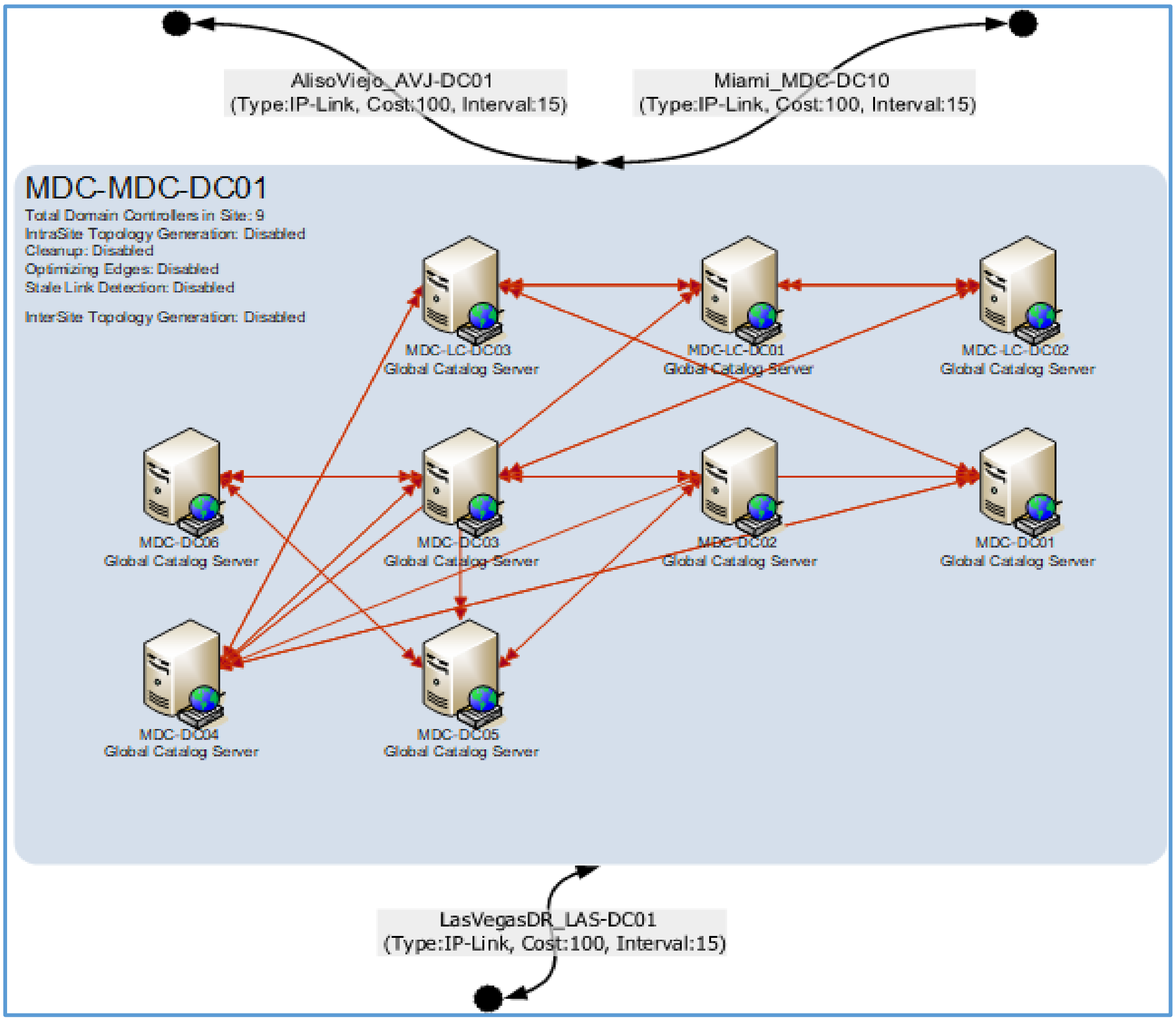

Figure 4: Lennar Intrasite Topology

The above figure is of the MDC-MDC-DC01 intrasite for Lennar.Lennarcorp.com and it shows the replication links between domain controllers that coincide in the site. This also shows the site links it has with the other sites that the domain controller resides in. Shown are the site links MDC-MDC-DC10 has with AlisoViejo_AVJ-DC01, LasVegasDR_LAS-DC01, and Miami_MDC-DC10.

Intersite Replication Topology

Inter-site replication takes place between sites and uses either RPC over IP or SMTP to replicate the data. Inter-site replication has to be manually configured and occurs between two domain controllers that are so-called bridgeheads. This role is assigned to at least one domain controller within a site. It is only these bridgeheads that replicate data with domain controllers in different domains by performing inter-site replication with its partners and packets are compressed to save bandwidth.

Click to view the pictorial representation - Lennar Intersite Topology.

RELATED DOCUMENT

- Active Directory Domain Services – Windows applications: https://docs.microsoft.com/en-us/windows/win32/ad/active-directory-domain-services
- Microsoft Article: https://support.microsoft.com/en-us/help/223346/fsmo-placement-and-optimization-on-active-directory-domain-controllers